

A TRANSFORMATION FOR SOLVING A PATTERN SEQUENCING PROBLEM IN THE WOOD CUT INDUSTRY

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Abstract

Consider a wood cutting setting where different panels have to be cut from large boards. Each cut panel size is put into a stack which remains opened until the last panel of that size is cut. The minimization of open stack problem (MOSP) is the problem of sequencing the cutting patterns in order to minimize the maximum number of open stacks during the production run. It is shown that an optimal solution to any MOSP can be obtained from an optimal solution of a corresponding MOSP where all patterns have at most two different panel types.

Keywords: cutting stock, pattern sequencing, pattern allocation, graph traversing

Resumo

Considere uma indústria de corte de madeira onde painéis de tamanhos diferentes são cortados de grandes chapas. Cada tamanho diferente de painel cortado abre uma pilha nova que permanece aberta até que o último painel daquele tamanho seja cortado. O problema de minimização de pilhas abertas (PMPA) consiste em determinar um sequenciamento dos padrões de corte de modo a minimizar o número máximo de pilhas abertas durante o processo de corte. Mostra-se que uma solução ótima para qualquer PMPA pode ser obtido de uma solução ótima de um outro PMPA correspondente onde todos os padrões tem no máximo dois tamanhos diferentes de painéis.

Palavras-chave: corte de estoque, sequenciamento de padrões, percorrimto em grafos

1. INTRODUCTION

Consider, for instance, a wood cutting setting where different panel sizes are to be cut from large boards. Assume we are given a set of patterns to be cut. The minimization of open stacks problem (MOSP) (see Yanasse [8]) deals with the sequencing of the patterns in order to minimize the maximum number of incomplete stacks. A stack is opened for each different panel type cut and it is closed only after the last piece of that type is cut. Closed stacks can be removed prior to the cutting of the next pattern. Consider, for instance, the set of patterns shown in Table 1.

Table 1

Patterns	Order Panels
P ₁	a b d
P ₂	b d e f
P ₃	c d
P ₄	a c e
P ₅	a f
P ₆	e f

A possible sequence of the patterns is P₁ P₂ P₃ P₄ P₅ P₆. This would result in the situation described in Table 2.

Table 2

Pattern sequenced	open panels	closed panels	number of open stacks
P ₁	a b d		3
P ₂	a b d e f	finished b	5
P ₃	a c d e f	finished d	5
P ₄	a c e f	finished c	4
P ₅	a e f	finished a	3
P ₆	e f	finished e and f	2

In some production settings MOSP is the problem of sequencing the production jobs in order to minimize the number of open orders of the customers. This has obvious implications on inventory costs, material handling, incoming payments, date of delivery of goods to clients, transportation costs, etc.

The MOSP is addressed in Yanasse [7], [8] where a non-trivial mathematical formulation, a branch and bound algorithm and a greedy heuristic algorithm are suggested. Relations of the MOSP with the minimization of order spread problem (see Garey et al. [2], Madsen [5]), and the minimization of the number of tool switches problem (see Tang and Denardo [6]) are also discussed. Yuen [10], [11] has presented, implemented and tested a few

heuristics for this problem. Other pattern sequencing problems treated in the literature [2], [5] seek different objectives.

Lins [4] proposes a heuristic method to solve the MOSP for the particular case where there are at most 2 different panel types per pattern and the graph associating the nodes to orders and arcs to patterns is a 1-tree, i.e. a tree with an additional arc. Yanasse [9] proposes polynomial algorithms for the 1-tree case and other graphs with very special topologies.

The general MOSP seems to be difficult to solve. We show that an optimal solution to *any* MOSP can be obtained by solving a corresponding MOSP where all patterns have at most two panel types. The corresponding MOSP can be obtained in polynomial computational time, hence, solving general MOSPs is as difficult as solving "simpler" ones where all patterns have at most two panel types.

In section 2 we point out some characteristics of the solutions of the MOSP. In section 3 we focus on the "particular" case where all patterns have at most 2 different panel types. We define the MOSP as a graph problem, in a similar way as done in [2] with the minimization of order spread problem. We then present a few observations regarding properties of an optimal solution to the MOSP in connection with the characteristics of the graph associated with it. In section 4 we present our main result, which is the possibility of solving any MOSP by means of another MOSP having at most two different panel types per pattern. Finally, in section 5 we make a few concluding remarks.

2. PROPERTIES OF SOLUTIONS TO THE MOSP

The identification of properties concerning the solutions of a problem is of interest for it may limit one's search for an optimal solution. A few properties of the MOSP were identified and are presented next.

Proposition 1: Consider a sequence $P_1 P_2 \dots P_n$ corresponding to a possible sequencing of the patterns of a MOSP. Let L_1 be the list of open stacks following this sequence. The reverse order sequence of patterns $P_n \dots P_2 P_1$ produces the same open stacks in reverse order, that is, list L_1 in reverse order.

Proof of Proposition 1: By contradiction. Suppose that the reverse order sequence does not produce the same open stacks at some instant. Then following the original sequence order we show that a different set of open stacks could not be produced.

To illustrate *Proposition 1*, take, for instance, the patterns presented in Table 1. A possible sequence of the patterns is $P_1 P_2 P_3 P_4 P_5 P_6$ and, in Table 2, column 2, we have the list of open stacks following this sequence. If we follow the reverse sequence $P_6 P_5 P_4 P_3 P_2 P_1$ we obtain the situation described in Table 3. Compare column 2 of Tables 2 and 3 and observe that the open stacks formed are the same in both cases but in reverse order.

The next result follows directly from *Proposition 1*:

Corollary 1: Any MOSP having more than one pattern to sequence has multiple optimal solutions.

Table 3

Pattern sequenced	open panels	closed panels	number of open stacks
P ₆	e f		2
P ₅	a e f		3
P ₄	a c e f		4
P ₃	a c d e f	finished c	5
P ₂	a b d e f	finished e and f	5
P ₁	a b d	finished a, b and d	3

Proposition 1 was first presented in [1] and has been used recently by Yuen and Richardson [12].

The next proposition relates the optimal solutions of two related MOSPs, where the first MOSP contains the set of patterns of the second problem.

Proposition 2: Suppose we have a MOSP, $PROB_1$, with N patterns to be sequenced. Suppose we have a second problem, $PROB_2$, with the same N patterns as problem $PROB_1$ plus a few more patterns. Then the optimal solution value of $PROB_2$ is greater than or equal to the optimal solution value of $PROB_1$.

Proof of Proposition 2: Trivial. Take the optimal solution of $PROB_2$ and obtain a feasible solution to $PROB_1$ by simply ignoring, in the sequence, the patterns that belong only to $PROB_2$.

Proposition 2 allows us to construct solutions to a MOSP by combining solutions of smaller sized, hopefully simpler, MOSP problems corresponding to subproblems (subset of patterns) of the original problem. *Propositions 1 and 2* give indications that a MOSP is likely to have an increase in the number of alternative optimal solutions as its size (number of patterns to sequence) increases. It also provides a possible way of determining lower bounds for the optimal value of a MOSP (by solving simpler MOSP problems).

The results described so far are valid for any MOSP. In the next section, we focus on properties of particular MOSPs where every pattern has at most two different panel types.

3. THE CASE WITH AT MOST TWO PANEL TYPES PER PATTERN

In [7], [8], relationships between the MOSP and the minimization of order spread problem (MORP) were discussed. The MORP is concerned with the reduction of the “lifetime” of an open stack, called its spread. The objective in this case is to maintain a stack open as little as possible. In order to achieve this goal, patterns that contain panels of the same type must be sequenced as close as possible to each other.

Consider the problem data set given in Table 1. Panel type a, for instance, appears in patterns P_1 , P_4 and P_5 . Hence, if we are interested in reducing the lifetime of the stack of panel type a once it is opened, then these patterns must be sequenced as close as possible to each other. If the lifetime of a stack is measured in terms of the number of different patterns

sequenced, then, the best we could do for panel a, for instance is 2, since this panel appears in 3 distinct patterns. In sequence $P_1 P_2 P_3 P_4 P_5 P_6$, the order spread of panel a is 4, since there are four different patterns since the instant panel a is opened (pattern P_1) until it is concluded (pattern P_5).

By maintaining stacks open as little as possible, the desired objective of minimizing the maximum number of open stacks perhaps may be achieved. Although these problems seem to be closely related, they are not equivalent. This can be seen by a simple example. Take, for instance, the set of data presented in Table 4.

Table 4

Patterns	Order Panels
P_1	a b
P_2	b c
P_3	c
P_4	a d e
P_5	d e f

An optimal sequencing for MOSP is, for instance, $P_1 P_2 P_3 P_4 P_5$, with a maximum of three open stacks. The maximum order spread associated with this sequence is 3 since panel a is opened in the first pattern and is closed just before the last pattern is sequenced. On the other hand, the sequence $P_1 P_4 P_2 P_5 P_3$ has an associated maximum order spread of 2 and is optimal for MORP. The maximum number of open stacks associated with this sequence is 4 and is not optimal for MOSP.

We add here a few additional remarks relating both problems. The MORP can be put as a graph problem (see Garey et alii [2]), where there is a vertex for each pattern and vertices are adjacent to each other if and only if the corresponding patterns have at least one order panel in common. Consider the problem data set given in Table 1.

The MORP graph representation of this data set is presented in Figure 1.

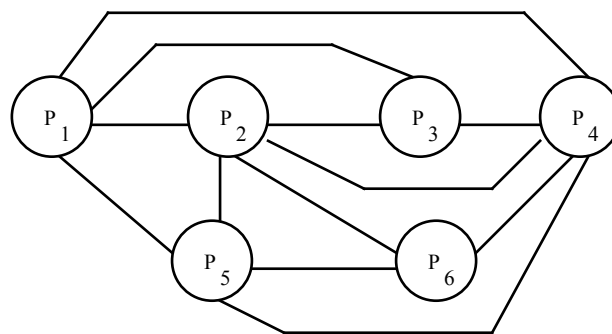


Figure 1

Graph representing MORP of data set given in Table 1

Let G be a graph with a set of vertices V . One can define (see Garey et alii [2]) a layout

of G as a mapping $f: V \rightarrow \mathbb{Z}^+$ or, equivalently, a layout may be seen as a ordered sequence of vertices with each vertex appearing exactly once. The correspondence between these two definitions is simply that $f(v) = k$ if and only if v is the k -th vertex in the sequence. The bandwidth of a layout f is defined as:

$$\text{bandwidth}(f) = \max \{|f(u) - f(v)| : \text{for all } u \text{ and } v \text{ that are adjacent vertices in } G\}.$$

The bandwidth of a graph G (or MORP as a graph problem) is:

$$\text{bandwidth}(G) = \min \{\text{bandwidth}(f) : \text{for all } f, \text{ layout of } G\}.$$

Since MOSP and MORP seem to be closely related, the natural question that arises is whether the MOSP can be also represented as a problem in a graph. The graph representation presented for the MORP, apparently, is not appropriate for the MOSP since the indication of how many stacks will be opened when sequencing the patterns is lost.

A graph representation of the MOSP is possible [4] for the case where there are at most two different orders (panel types) per job (pattern). The graph associates nodes to orders and arcs to patterns. Consider the problem data set given in Table 5.

The MOSP graph associated with the data set of Table 5 is shown in Figure 2a. In Figure 2b the graph associated to the MORP using the same data set is also presented.

Table 5

Patterns	Order Panels
P ₁	a b
P ₂	b c
P ₃	c d
P ₄	d e
P ₅	c f

MOSPs whose associated graphs are trees can be solved in polynomial time (see Yanasse [9]). This is not true for MORPs. It is known [2] that for MORP, even if the graph associated with it is a tree with all vertices with degree smaller or equal to 3, the decision whether it is possible or not to find a sequence with bandwidth less or equal to k , given k , is NP-hard (to test whether the bandwidth is equal 1 or 2 can be done in polynomial time).

For the remaining of this section we consider only MOSPs where all patterns have at most two different panel sizes per pattern. These MOSPs can be seen as a problem of traversing graphs.

From now onwards, since the graph associated to MOSP associates nodes to panels and arcs to patterns, we will use these associated terms without distinction.

Let $\mathbf{P}$ be a MOSP and let $G(V, E)$ be the graph resulting from $\mathbf{P}$. V and E are the set of nodes and arcs in the graph, respectively.

There are a few observations that can be proven for problems $\mathbf{P}$ of this type. Recall that the number of open stacks are counted *after* a pattern is completely cut and *before* any finished stack is removed. A stack is finished when all the requirements for its particular panel type is completed.

Proposition 3 links the MOSP with the maximum clique problem in a graph (see Garey and Johnson [3]). We can state:

Corollary 2: If in an graph $G(V, E)$ corresponding to a MOSP $\mathbf{P}$ the maximum clique is of size n , then in an optimal solution to $\mathbf{P}$, the maximum number of open stacks during the production run is at least n .

Proof of Corollary 2: Follows immediately from *propositions 3* and *2*.

Take, for instance, the graph of Figure 4. We can easily identify a maximum clique of size 4 in this graph. If we consider just this clique and delete all other arcs and nodes of the original graph, the resulting graph has all nodes with degree 3. Hence, by proposition 3, we will get at least 4 open stacks at some time during cutting. Also, by proposition 2, we know that this is a lower bound on the number of open stacks for the original graph.

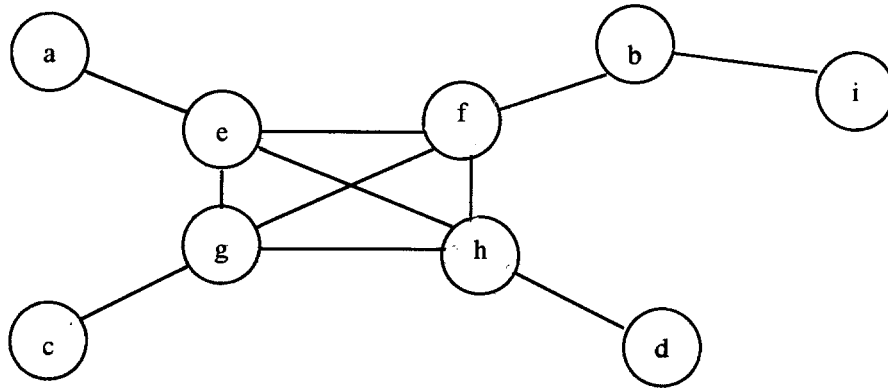


Figure 4
Instance of a graph to illustrate corollary 2

Therefore, by *corollary 2* we can conclude that if an optimal solution value to a MOSP is n then there is no clique of size $n+1$ or more in the graph. Unfortunately, we cannot guarantee that there will be a clique of size n or less in the graph. It is possible that a graph $G(V, E)$ corresponding to a problem $\mathbf{P}$, has no cycles at all (a tree) and $\mathbf{P}$ might have a large number of open stacks in an optimal solution. It seems that the MOSP is NP-hard but this is still an open question.

A natural question that is posed is whether it is possible to identify cliques of a graph from an optimal sequencing of the corresponding MOSP. The following proposition might be of some help in this aspect.

Proposition 4: Suppose that in the graph $G(V, E)$ corresponding to a MOSP there is a clique of size n , $n \geq 4$. In this case, any sequencing of patterns will have a subsequence of at least 4 consecutive patterns, where the corresponding resulting number of open stacks will be at least $n-1$.

Proof of proposition 4: Consider the subgraph corresponding to the clique of size n . During the sequencing of the patterns we will certainly reach a stage where there are $n-2$ open stacks corresponding to vertices of the clique. When any pattern corresponding to an arc connecting any of these $n-2$ vertices to the $(n-1)^{\text{th}}$ vertex of the clique is sequenced and cut, there will be $n-1$ open stacks left corresponding to vertices of the clique, since the patterns corresponding to the arcs connecting these $n-2$ vertices with the n^{th} vertex of the clique were not sequenced yet and, hence, they cannot be finished. When any of the patterns corresponding to arcs connecting to the n^{th} vertex of the clique is sequenced, the panel size corresponding to the n^{th} vertex will open; some other panel size corresponding to a vertex of the clique may finish but still we have again at least n open stacks. If one of the panel sizes is finished, it has got to be one corresponding to one of the initially open $n-2$ vertices of the clique for there are still at least $n-2$ patterns to sequence corresponding to arcs incident to the $(n-1)^{\text{th}}$ vertex and also, at least $n-2$ patterns to sequence corresponding to arcs incident to the n^{th} vertex. Therefore, there are at least two arcs incident to each one of the remaining vertices of the clique whose corresponding patterns need to be sequenced before any one of the panel sizes corresponding to these vertices is finished. The proposition then follows.

We next focus our attention on the general MOSP since, in practical problems, the patterns are not restricted to have at most two different panel sizes. We will relate the general MOSP with a "simpler" one having at most two panels per pattern.

4. ON SOLVING GENERAL MOSPs

The following interesting result allows us to concentrate our efforts on the development of algorithms for MOSPs with at most two panels per pattern.

Proposition 5: An optimal solution for any pattern sequencing problem can be obtained from an optimal solution of a pattern sequencing problem with at most two panels per pattern, appropriately built from the data of the original problem.

This proposition indicates that the "at most two panels per pattern" problem represents, in fact, no loss of generality.

The original problem and the at most two panels per pattern problem obtained from the original problem are not equivalent in the sense that a feasible solution to one problem has always a corresponding (unique) feasible solution to the other problem. However, given an optimal solution to one problem, we can find an equivalent optimal solution to the other (and vice-versa) in polynomial time.

Given a general MOSP, the graphical representation of its "simpler" problem is obtained by simply introducing a clique in the graph in correspondence to each pattern in the original problem. Hence, if a pattern has k panel types, there will be a clique of size k corresponding to this pattern in the graphical representation. The k nodes of this clique is obviously the nodes corresponding to each one of the panels in the patterns. For example, if we have the patterns given in Table 1, we would have the graph illustrated in Figure 5.

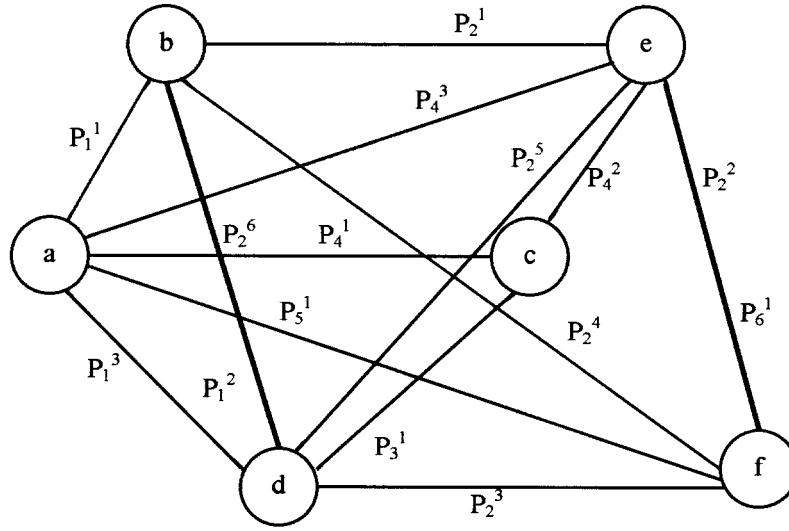


Figure 5
Graph representation of problem data given in Table 1

Note that there will be patterns that coincide in the modified two panels per pattern problem. For instance, in the example presented, patterns P_1^2 and P_2^6 are the same pattern. Observe also that the modified problem can be obtained in polynomial time from the original problem.

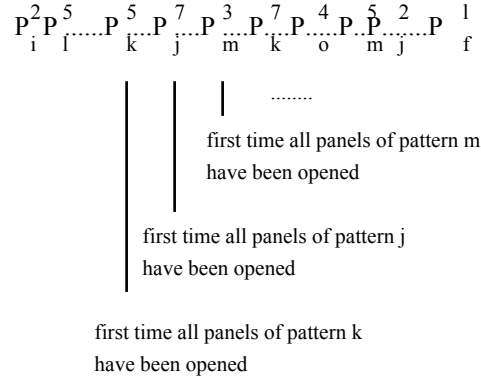
Proof of Proposition 5: Let $P_1, P_2, \dots, P_n$ be the original patterns of our initial MOSP $\mathbf{P}$. Let $P_1^1, P_1^2, \dots, P_1^{m_1}, P_2^1, P_2^2, \dots, P_2^{m_2}, \dots, P_n^1, P_n^2, \dots, P_n^{m_n}$ be the patterns of our surrogate "at most two panels per pattern" problem, obtained as explained previously, where $m_1, m_2, \dots, m_n$ are the total number of patterns in the new problem in place of $P_1, P_2, \dots, P_n$, respectively. If s_i is the total number of panel types in pattern P_i , $i = 1, 2, \dots, n$, then m_i is equal to $s_i(s_i-1)/2$. Let us refer to this surrogate problem as $\mathbf{P}_{\text{mod}}$. We will prove the proposition by showing that an optimal solution to $\mathbf{P}$ has an equivalent optimal solution to $\mathbf{P}_{\text{mod}}$ and vice-versa.

Let S_1 be a solution to $\mathbf{P}$. A solution S_2 to $\mathbf{P}_{\text{mod}}$ in correspondence to S_1 of $\mathbf{P}$ can be constructed in the following way: "for each pattern in S_1 , sequence all arcs of the clique corresponding to it; arcs that have been sequenced before are, obviously, not repeated".

So, if $S_1 = P_1 P_j \dots P_k$ is the solution for $\mathbf{P}$, then the corresponding solution in $\mathbf{P}_{\text{mod}}$ is of the form $P_1^1 \dots P_1^{m_1} P_j^1 \dots P_j^{m_j} \dots P_k^1 \dots P_k^{m_k}$. It is easy to see that the maximum number of open stacks resulting in both problems is the same. Consider, one at a time, the successive cliques whose arcs are being sequenced in $\mathbf{P}_{\text{mod}}$. Due to *proposition 3*, all the nodes belonging to the clique are going to be simultaneously opened some time during the sequencing. Therefore, the same stacks opened just after sequencing each pattern in S_1 are also going to be opened using the corresponding set of patterns proposed in $\mathbf{P}_{\text{mod}}$ (not necessarily after all arcs of the corresponding clique are sequenced but surely sometime during their sequencing). Observe that no additional stacks besides those corresponding to the clique under consideration are opened (since the arcs being sequenced are incident only to nodes of the clique under consideration). All the stacks that are closed or remain opened after a pattern in S_1 is

sequenced will also be closed or remain opened after all the arcs of the corresponding clique are sequenced in $\mathbf{P}_{\text{mod}}$. Hence, the maximum number of stacks in both problems is the same.

Consider now a solution S_2 to problem $\mathbf{P}_{\text{mod}}$. A solution S_1 to problem $\mathbf{P}$ can be obtained as follows. Consider after each pattern sequenced in S_2 , the panels that were already opened. The point where all panel sizes of an original pattern (a pattern of problem $\mathbf{P}$) have been opened indicates the position of this pattern in the sequence S_1 . Suppose solution $\mathbf{P}_{\text{mod}}$ is of the form



The corresponding solution to $\mathbf{P}$ is $P_k P_j P_m \dots P_r$.

Observe that the maximum number of open stacks obtained with sequence S_1 in problem $\mathbf{P}$ is less or equal to the maximum number of open stacks obtained with sequence S_2 in problem $\mathbf{P}_{\text{mod}}$. This is so by the following argument. Consider the solution S_2 , $P_i^2 P_f^5 \dots P_k^5 P_j^7 \dots P_m^3 P_k^7 P_o^4 P_m^5 P_j^2 \dots P_f^1$ where, for instance, after each pattern P_k^j sequenced, $j = 1, 2, \dots, n$ we identified that the pattern $P_k(j)$ of the original problem is to be sequenced. Let S_2^j denote a subsequence of S_2 formed from the start of sequence S_2 up to pattern P_k^j . Similarly, let S_1^j denote a subsequence of S_1 formed from the start of sequence S_1 up to pattern $P_k(j)$. Let m_2^j be the number of open stacks at the instant just after P_k^j using sequence S_2 and m_1^j be the number of open stacks at the instant just after $P_k(j)$ using sequence S_1 . Then $m_2^j \geq m_1^j$, $j = 1, 2, \dots, n$ since, otherwise, suppose for some g , $m_2^g < m_1^g$. This means that at some point P_h^g in sequence S_2 , there is at least one panel, say r , that has been finished using S_2^g but has not been finished using S_1^g . If that is so, in S_2^g there should be at least one pattern in the sequence, say P_o^t , corresponding to a pattern P_t not yet sequenced in S_1^g . Since P_t has not been sequenced in S_1^g , it has at least another panel p which has not been opened yet. But r also belongs to P_t and, by construction, r should have an arc linking it to p . Hence, r could not be finished.

The solutions we considered so far to problems $\mathbf{P}$ and $\mathbf{P}_{\text{mod}}$ were any. In particular, if we choose $S_1 = S_1^*$, an optimal solution to $\mathbf{P}$, we obtain a solution S_2 , in correspondence to S_1^* , which is optimal to $\mathbf{P}_{\text{mod}}$. Similarly, if we choose $S_2 = S_2^*$, an optimal solution to $\mathbf{P}_{\text{mod}}$, we obtain a solution S_1 , in correspondence to S_2^* , which is optimal to $\mathbf{P}$. The optimality of the solutions can be proved by contradiction. □

So, the general graph MOSP represents any possible case of the pattern sequencing problem, irrespective of the number of panels per pattern in the problem. Observe that many different instances of the MOSP with different set of patterns might have the same "at most two panels per pattern" representation graph. Therefore, they will have the same optimal

value.

Consider the graph representation of the problem data given in Table 1 (Figure 5).

Other different problems that would result in the same graph representation are given, for instance, in Tables 6 and 7.

Table 6

Patterns	Order Panels
P ₁	a b d f
P ₂	a c d e
P ₃	a b d e
P ₄	c e
P ₅	b e f

Table 7

Patterns	Order Panels
P ₁	a b d e f
P ₂	a c d e

Recalling the relationship between MOSP and the maximum clique problem, we can state that finding the maximum clique of a graph is equivalent to finding a related pattern sequencing problem with the condition that it contains a pattern having the largest possible number of panel types in it. Using the previous example, in Table 6, the patterns that have the largest number of panel types are P₁, P₂ and P₃ with 4 panel types. In Table 7, we have another related pattern sequencing problem with a pattern having 5 panel types. Since the maximum clique of the graph corresponding to these problem is 5 (see Figure 5) there is no pattern sequencing related problem with at least one pattern having more than 5 panel types.

Proposition 5 also provides additional support to the conjecture that MOSP with at most two panel types per pattern is NP-hard. Recall that in [7], [8] the relationship of the MOSP and the minimization of tool switches problem in a flexible manufacturing system is presented and this latter problem seems to be also a computationally difficult problem to solve (see Crama [1]).

It may be worth observing that the graph representation of MOSP suggested can also be described as an hypergraph.

5. SUMMARY AND CONCLUSIONS

We showed that an optimal solution for any of our pattern sequencing problem can be obtained from an optimal solution of a "simplified" pattern sequencing problem with at most two panel types per pattern. This is a relevant result since any pattern sequencing problem can be put now as a problem in a graph. The motivation in analysing a pattern sequencing problem as a graph problem has, therefore, greatly increased since, previously, these cases were supposed to represent only very special and limited practical instances (see, for instance, Lins [4] and Yanasse [9]).

In [7], [8] a branch and bound method for solving general MOSPs is presented. With the result of *proposition 5*, other alternative schemes can be suggested. By using *propositions 1*, *2* and *5*, we can reduce the search in the above mentioned branch and bound method. Once the graph corresponding to the original problem is determined, it can be subdivided into

simpler subgraphs. Lower bounds for the original problem can be determined by the optimal solutions of the subgraphs and possibly by using *proposition 3* too.

The relation of the MOSP with the clique problem as shown in *proposition 4* and *corollary 2* adds to the conjecture presented in [7], [8] that the general MOSP is computationally hard to solve.

The MOSP seems to be related closely to the MORP and to the minimization of the number of tool switches problem in a flexible manufacturing system, hence, it may be worth mentioning that many of the propositions presented can be extended similarly to these problems.

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